

praveenkumar@gmail: ~

GNU nano 8.7.1

Exp27.cpp

```
#include <systemc.h>

SC_MODULE(Binary_semaphore) {

    int sem;
    sc_event ev;

    void task1() {
        while(true) {
            wait(1, SC_SEC);
            if(sem == 1) {
                sem = 0;
                cout << "Task1 Entered Critical Section at " << sc_time_stamp() << endl;
                wait(2, SC_SEC);
                sem = 1;
                ev.notify();
            } else {
                wait(ev);
            }
        }
    }

    void task2() {
        while(true) {

```

[Wrote 53 lines]

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```
praveenkumar@gmail:~$ nano Exp27.cpp
praveenkumar@gmail:~$ g++ -o Exp27 Exp27.cpp -lsystemc
praveenkumar@gmail:~$ ./Exp27
```

```
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
```

```
Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
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```